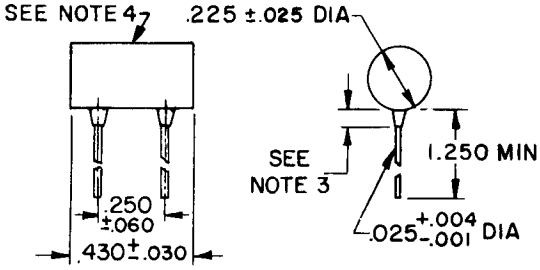
MILITARY SPECIFICATION SHEET

# CAPACITORS, FIXED, CERAMIC DIELECTRIC (GENERAL PURPOSE), STYLE CK22

The complete requirements for procuring the capacitors described herein shall consist of this document and the issue in effect of Specification MIL-C-11015.




| INCHES | MM |
|---|---|
| .001 | .03 |
| .004 | .10 |
| .025 | .64 |
| .030 | .76 |
| .060 | 1.52 |
| .225 | 5.72 |
| .250 | 6.35 |
| .430 | 10.92 |
| 1.250 | 31.75 |

NOTES:
1. All dimensions in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Insulating coating shall not extend more than .125 (3.18 mm) along lead wires measured from body of capacitor.
4. The beginning of the type designation shall be adjacent to the end having the inner-electrode terminal.

**NOT FOR NEW DESIGN**
**FOR REPLACEMENT PURPOSES ONLY**

REQUIREMENTS:

Design and construction:
Dimensions and configuration - See above figure.
Case type - Tubular with radial leads.
Capacitance value - See table I.
Capacitance tolerance - ±10% (K) or ±20% (M) as shown in table I.
Rated temperature - -55° to +85° C.

Dielectric withstanding voltage:
Dielectric: Method 301 of MIL-STD-202.
Test voltage - 250% of rated voltage.
Body insulation: Shall be able to withstand 1,300 volts for 5 ± 1 seconds between leads and a V-block which shall extend beyond the ends of the capacitor body.

Barometric pressure (reduced): Method 105 of MIL-STD-202. 0.82 inch of mercury (80,000 ft).
Test potential - 150% of rated voltage.

Insulation resistance (IR): Method 302 of MIL-STD-202, condition B. 20,000 megohms, min.

Capacitance: Method 305 of MIL-STD-202.
Test frequency:
100 pf or less - 1 MHz ± 1 kHz.
Over 100 pf - 1 kHz ± 100 Hz.
Test voltage - 1.0 ± 0.2 volt rms.

Dissipation factor (DF): 1.5%, max.

Shock, medium impact: Not applicable.

Vibration, high frequency: Method 204 of MIL-STD-202, condition B (15 G's).

Temperature cycling and immersion:
Temperature cycling: Method 102 of MIL-STD-202, condition D.
Immersion: Method 104 of MIL-STD-202, condition B.
IR - 3,000 megohms, min.

(C) No technical changes FSC 5910


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TABLE I. Style CK22 characteristics.

| Type designation 1/ | Rated voltage | Rated temperature and voltage-temperature limits | Capacitance | Capacitance tolerance |
|---|---|---|---|---|
| | volts, dc | | pf | |
| CK22AX120K | 500 | AX | 12 | K |
| CK22AX150K | 500 | AX | 15 | K |
| CK22AX180K | 500 | AX | 18 | K |
| CK22AX220K | 500 | AX | 22 | K |
| CK22AX270K | 500 | AX | 27 | K |
| CK22AX330K | 500 | AX | 33 | K |
| CK22AX390K | 500 | AX | 39 | K |
| CK22AX470K | 500 | AX | 47 | K |
| CK22AX560K | 500 | AX | 56 | K |
| CK22AX680- | 500 | AX | 68 | K, M |
| CK22AX820K | 500 | AX | 82 | K |
| CK22AX101- | 500 | AX | 100 | K, M |
| CK22AX121K | 500 | AX | 120 | K |
| CK22AX151- | 500 | AX | 150 | K, M |
| CK22AX181K | 500 | AX | 180 | K |
| CK22AX221- | 500 | AX | 220 | K, M |
| CK22AX271K | 500 | AX | 270 | K |
| CK22AX331- | 500 | AX | 330 | K, M |
| CK22AX391K | 500 | AX | 390 | K |
| CK22AX471- | 500 | AX | 470 | K, M |
| CK22AX561K | 500 | AX | 560 | K |
| CK22AX681- | 500 | AX | 680 | K, M |

1/ Where applicable, the complete type designation will include an additional symbol to indicate capacitance tolerance.

Salt spray (corrosion): Not applicable.

Terminal strength: In accordance with MIL-C-11015, 5 pounds applied.

Moisture resistance: Method 106 of MIL-STD-202.
IR - 3,000 megohms, min.

Solderability: Method 208 of MIL-STD-202.
2 terminals.

Resistance to soldering heat: Not applicable.

Voltage-temperature limits: In accordance with MIL-C-11015.

Life (at elevated ambient temperature): Method 108 of MIL-STD-202. 1,000 hours (for qualification); 250 hours (performance check); 750 hours (continuation test).
Test potential - 200% of rated voltage.
IR - 500 megohms, min (at 85° C);
10,000 megohms, min (at 25° C).
DF - 3%, max (at 25° C).

Marking: In accordance with MIL-C-11015.



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Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL, MI
Navy - EC, AS, OS, SH
Air Force - 11, 17, 85
DSA - ES

User activities:
Army -
Navy - MC
Air Force - 19

Preparing activity:
Navy - EC

(Project 5910-0829-3)

Code "C"



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